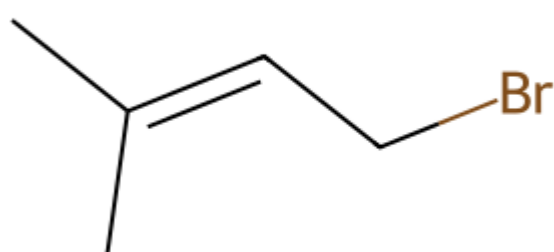


```
{'analysis_time': 0.0,  
'first_solution_time': 0.115017,  
'num_calls_rxn_model': 0,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 0.32296388659642067,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 0.32296388659642067,  
'retro_star_value': 0.32296388659642067}
```

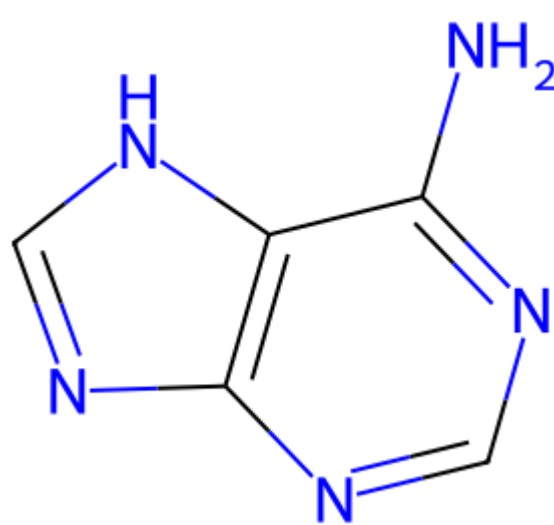


Reaction

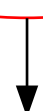
```
{'analysis_time': 0.115017,  
'first_solution_time': 0.203287,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 6.331141914678144,  
'retro_star_reaction_number': 6.331141914678144,  
'retro_star_rxn_cost': 3.912023005428146,  
'retro_star_value': 6.331141914678144}
```



```
{'analysis_time': 0.115017,  
'first_solution_time': 0.115017,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 6.331141914678144}
```

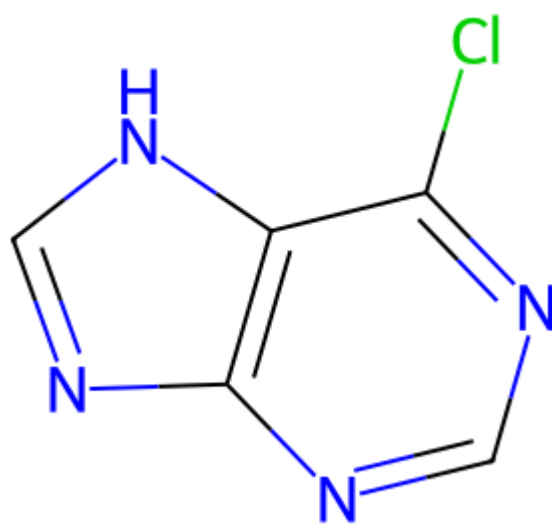


```
{'analysis_time': 0.115017,  
'first_solution_time': 0.203287,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 2.419118909249997,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 2.419118909249997,  
'retro_star_value': 6.331141914678144}
```



Reaction

```
{'analysis_time': 0.203287,  
'first_solution_time': 0.319729,  
'num_calls_rxn_model': 2,  
'num_calls_value_function': 35,  
'retro_star_min_cost': 4.975944054306325,  
'retro_star_reaction_number': 4.975944054306325,  
'retro_star_rxn_cost': 0.707246104939447,  
'retro_star_value': 8.887967059734471}
```



```
{'analysis_time': 0.203287,  
'first_solution_time': 0.203287,  
'num_calls_rxn_model': 2,  
'num_calls_value_function': 35,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 8.887967059734471}
```

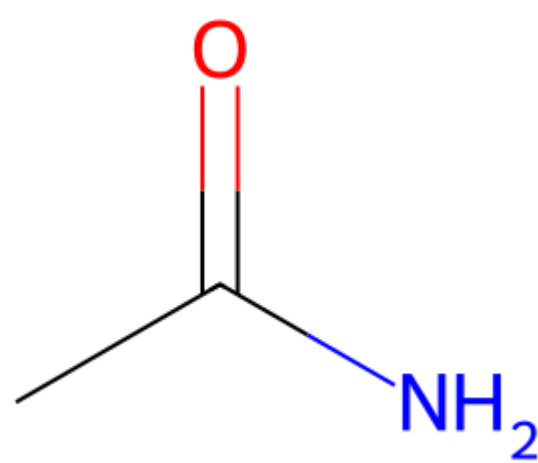


```
{'analysis_time': 0.203287,  
'first_solution_time': 0.319729,  
'num_calls_rxn_model': 2,  
'num_calls_value_function': 35,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 4.268697949366878,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 4.268697949366878,  
'retro_star_value': 8.887967059734471}
```



Reaction

```
{'analysis_time': 0.319729,  
'first_solution_time': 0.319729,  
'num_calls_rxn_model': 4,  
'num_calls_value_function': 81,  
'retro_star_min_cost': 5.521460917862246,  
'retro_star_reaction_number': 5.521460917862246,  
'retro_star_rxn_cost': 5.521460917862246,  
'retro_star_value': 10.14073002822984}
```



```
{'analysis_time': 0.319729,  
'first_solution_time': 0.319729,  
'num_calls_rxn_model': 4,  
'num_calls_value_function': 81,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 10.14073002822984}
```